

Compliance Calculation Book — VBF-T6R2-CALC-001

Sheet: Criteria

Criterion	Threshold	Result (DFN)	Unit
Volumetric energy density	950	1096	Wh/L
Voltage plateau (midpoint)	4.1	4.1301	V
Max temperature (4C, 45C amb.)	323.15	322.7	K
Lithium plating	False (min ap > 0)	False (min +0.0382)	V
SEI after 100 cycles	500	415.6	nm

Sheet: Derived

Quantity	Formula	Value	Unit
Energy	from 1C DFN discharge integral	26.518	Wh
Stack thickness	sum of layer thicknesses	235.6	um
Geometric area	electrode height x width	0.1027	m2
Volume	thickness x area	2.4196e-05	m3
ED volumetric	energy / volume (electrolyte excluded)	1096	Wh/L
Mass	sum(layer areal mass x area)	0.053772	kg
ED gravimetric	energy / mass	493.16	Wh/kg
DCR	(V(0) - V(10% t)) / I_1C	0.006866	ohm
Peak power density	V_OC^2 / (4 R_DC) / mass	12254.5	W/kg
SPMe cross-check ED	same formulas, SPM 1C	1106.3	Wh/L
SPMe cross-check midV	same formulas, SPM 1C	4.2202	V